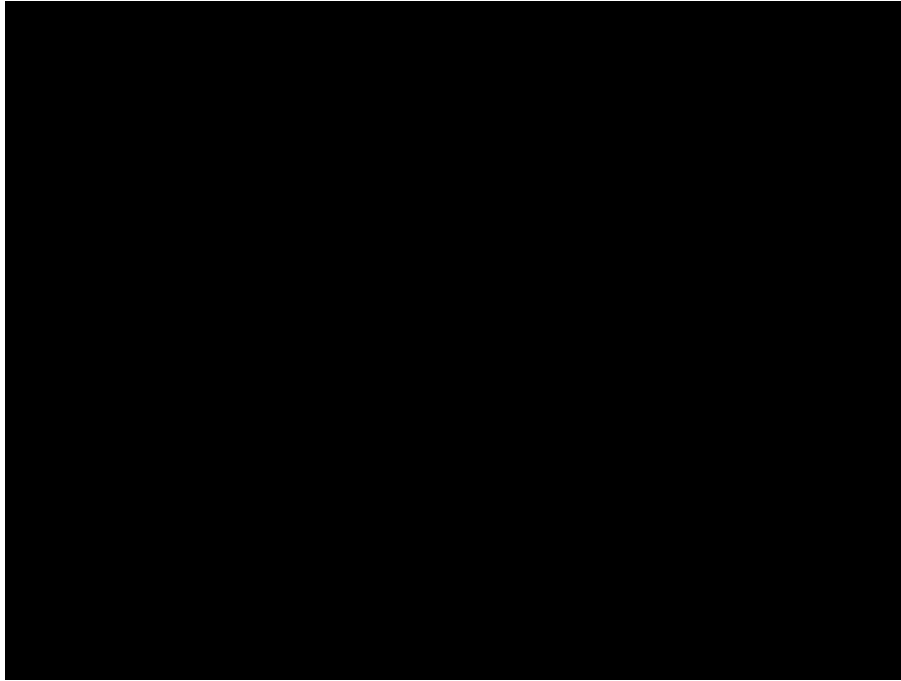

Moving Detection & Tracking

Sofía Lucía - up202200682

IMCV- FEUP

Input sequences characteristics



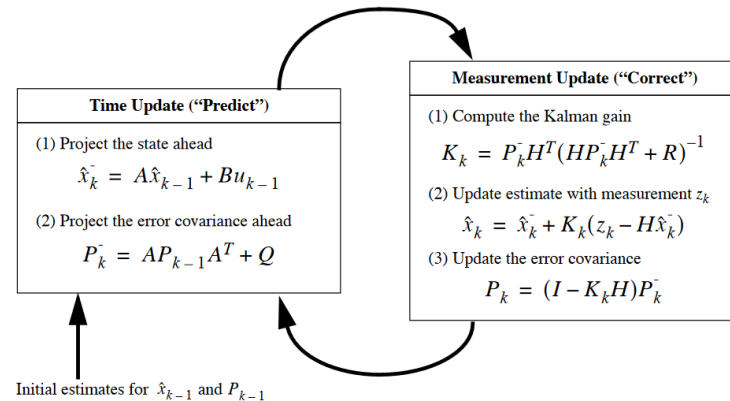
- VOT challenge dataset
- Name: surfer
- Frames: 841
- Dimensions: (360, 480, 3)

Kalman filter - Algorithm

The basic idea is that by using the prior knowledge of the state, the filter makes a forward projection state or predicts the next state.

The equations of the Kalman filter are divided into two groups:

- The **time update equations** are responsible for projecting the current state and error covariance estimates forward to obtain the a priori estimates for the next time step.
- The **measurement update equations** are responsible for obtaining the a posteriori estimates by incorporating a new measurement into the a priori estimates.



Kalman filter - Output



- sampling time (time for 1 cycle)=0.1
- acceleration in x-direction=1
- acceleration in y-direction=1, process noise magnitude=1
- std. dev. of the measurement in x-direction=0.1
- std. dev. of the measurement in y-direction=0.1



- sampling time (time for 1 cycle)=0.1
- acceleration in x-direction=1
- acceleration in y-direction=1, process noise magnitude=1
- std. dev. of the measurement in x-direction=0.3
- std. dev. of the measurement in y-direction=0.3

CamShift and MeanShift

Camshift is an advanced version of the **Mean Shift Algorithm**, but it can track the object even if it changes - **Dynamic Bounding Box**

Mean Shift:

- Works by defining a tracking window and shifting the center of the window to the centroid of the window.

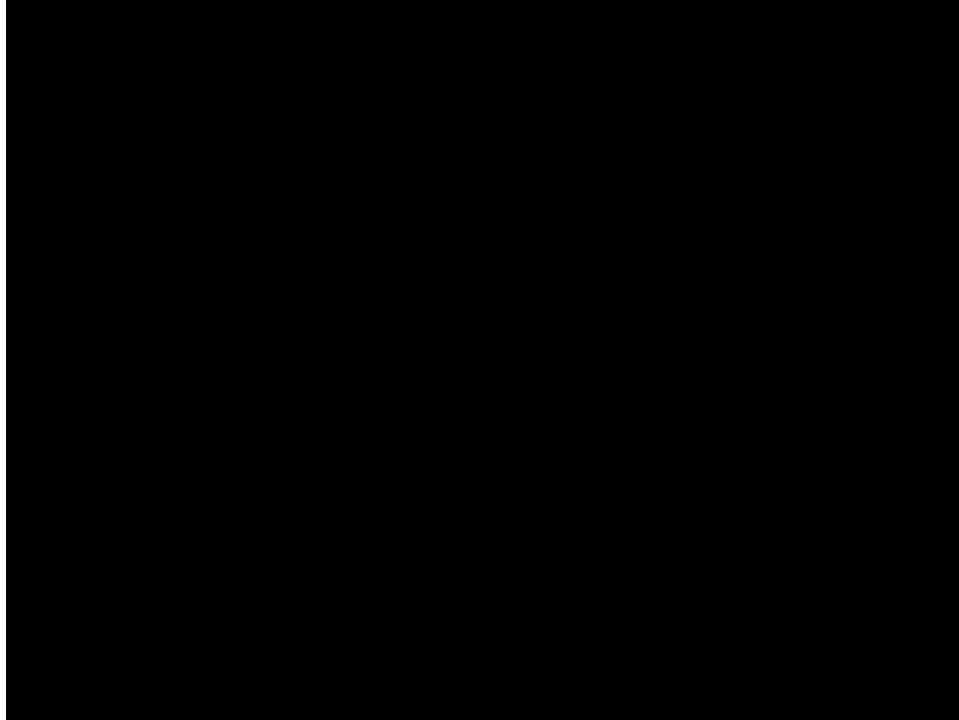


CamShift

1. Apply MeanShift
2. Update window size by the equation
3. Compute the orientation of the best fitting window
4. Re-apply Meanshift until convergence

$$s = 2 * \sqrt{\frac{M_{00}}{256}}$$

Results



CamShift

Advantages

- Easy to implement.
- Light-weight.

Disadvantages

- Sensitive to initialization
- Limited to single object tracking
- Dependent on color

Comparative

	Accuracy	Precision	IoU	Time (sec/frame)
Kalman filter (params_1)	0.991	0.914	0.847	0.001
Kalman filter (params_2)	0.983	0.835	0.738	0.001
CamShift	0.921	0.791	0.066	0.00058

Thank you!